

Within the specialized microenvironment of an enzyme's active site, which of the following amino acids has a side chain that can act as a nucleophile?

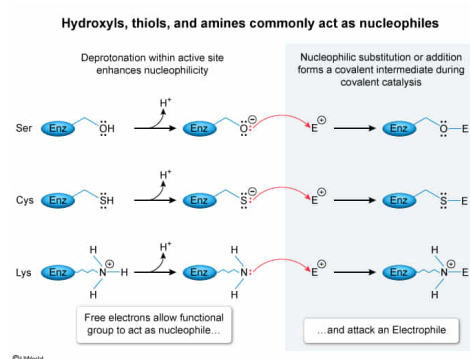
- I. Cysteine
- II. Lysine
- III. Serine

- ☐ A. II only [13%]
☐ B. III only [28%]
☐ C. I and II only [18%]
✓ ☒ D. I, II, and III [39%]

Correct

38% answered correctly

Explanation:



Enzymes can speed up chemical reactions by a variety of methods. One way is **covalent catalysis**, which occurs when **a temporary covalent bond is formed between the substrate and the enzyme itself**. These covalent bonds are commonly created through **nucleophilic substitution** or **addition** reactions through either an amino acid side chain or an **enzyme cofactor**. Several amino acids contain functional groups in their **side chains** that can serve as **weak nucleophiles** in aqueous solution, and the nucleophilicity of these amino acids is **enhanced even further in the active site** of the enzyme.

Hydroxyl oxygens have two free electron pairs available to seek and bond to an electrophile. Thus, the **hydroxyl groups of Ser, Thr, and Tyr** are nucleophiles, albeit weak ones. In the enzyme active site, however, nearby residues (eg, Asp, Glu, His) can accept a proton, deprotonating the hydroxyls into the **alkoxide (S, T)** or **phenoxide (Y)** form. With a formal negative charge, the oxides are much more strongly nucleophilic. Similarly, **the thiol of cysteine** is deprotonated to a thiolate. Thiolates are **even stronger nucleophiles** than oxides due to the **larger radius** and larger **polarizability** of the Period 3 sulfur atom. Therefore, cysteine and serine can act as nucleophiles (**Numbers I and III**). Serine proteases and cysteine proteases are common examples where these residues act as nucleophiles.

In isolation, the **amino functional group of lysine** would be protonated at a physiological pH of 7.4, would not have any free electrons, and could not be even a weak nucleophile. But as before, the active site structure can position residues nearby that deprotonate the functional group, freeing up a pair of electrons to attack a substrate electrophile. Although -NH_2 groups are still weak nucleophiles, **they can still be effective** in combination with enzymes' other catalytic mechanisms (eg, orientation of substrates, desolvation, binding energy). Therefore, lysine can also act as a nucleophile (**Number II**). Common examples of lysine acting as a nucleophile include ubiquitination and histone acetylation.

Educational objective:

Hydroxyls, thiols, and amines can serve as nucleophiles to form covalent bonds in nucleophilic substitution or addition reactions during covalent catalysis. Although these functional groups may be weak or inactive nucleophiles in isolation, other residues in the enzyme active site can deprotonate these groups to enhance

their nucleophilicity.

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